BIG DATA FRAMEWORK

"Predicting Cascading Failures in Critical Infrastructure Networks"

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INTRODUCTION

- Modern critical infrastructure is deeply interconnected — power, water, transportation, and telecom systems depend on each other to function, so they can no longer be studied or protected in isolation.

- A failure in one system can trigger failures in others — for example, a power outage can shut down water pumping stations, disrupt traffic signals, and take down telecom towers, creating a chain reaction across sectors.

- These cascading failures are hard to predict with traditional methods — the scale, speed, and complexity of interdependent networks make manual or rule-based monitoring insufficient for catching failures before they spread

- Big Data and AI offer a path to early, proactive detection — by modeling infrastructure as interconnected graphs and applying machine learning, it becomes possible to predict how and where a failure will propagate, enabling faster response and damage prevention

MOTIVATION

- Modern infrastructure (power, water, transport, telecom) is deeply interconnected - failures rarely stay isolated and often cascade further than the original fault (e.g., 2003 Northeast blackout, 2021 Texas grid failure)
- Current systems are largely reactive - failures are detected after the fact, not predicted in advance
- Big Data + AI (GNNs, ML) now make real-time cascade prediction feasible, even as climate change increases the frequency of compounding infrastructure stress
- Predictive models enable early warning, better resource allocation, and disaster prevention

SDG Alignment





LITERATURE REVIEW

S.No	Author, Title & Year	Objective	Methodology	Significance
1	A. Le, P. K. Huynh, O. P. Yadav, H. Pirim, C. Le, T. Q. Le — "Topology-Aware Spatio-Temporal Graph Transformer for Predicting Smart Grid Failures," RAMS 2026	Improve smart-grid resilience/predictive maintenance by forecasting next-day substation failures, addressing weak modeling of spatio-temporal-causal interdependencies and class imbalance in existing methods.	Proposed a Topology-Aware Spatio-Temporal Graph Transformer (ST-GT) that embeds physical transmission-network topology directly into the transformer's attention mechanism, jointly processing static topological descriptors and temporal Phasor Measurement Unit (PMU) sequences end-to-end. Evaluated via 5-fold cross-validation across 10 substations.	Achieved perfect recall (1.000) and F1 = 0.858, substantially beating XGBoost baselines (0.683); gives interpretable insight into failure-propagation pathways, supporting proactive, scalable grid maintenance.
2	N. Ye — "Hybrid Multimodal Knowledge Graph and Meta-Learning Framework for Accurate Dynamic RP-VRC Prediction in Renewable-Rich Power Systems," IEEE Access, 2025	predict dynamic reactive-power/voltage-related risk conditions (RP-VRC) in power systems with high renewable penetration.	combines a multimodal knowledge graph (fusing heterogeneous grid/weather/operational data sources) with a meta-learning framework to adapt quickly to changing renewable-generation conditions.	Targets accuracy/adaptability challenges renewables introduce to grid stability prediction; supports safer integration of variable renewable sources.
3	T. Zhang, D. Tang, P. Fan, Q. Wang, P. Wang — "A Probabilistic Graphical Model for Predicting Cascade Failures of Electric Vehicle Charging Networks Caused by Hurricanes," IEEE Trans. Smart Grid, 2025	Predict how hurricanes trigger cascading failures across EV charging networks, given their growing interdependence with the power grid	Builds a probabilistic graphical model (likely Bayesian-network style) capturing conditional dependencies between hurricane hazard variables, power-grid outages, and EV charging-station failures/cascades.	Addresses a newly critical infrastructure-resilience gap as EV adoption grows; informs hurricane-preparedness and charging-network hardening strategies

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4	T. Chen, Z. Guo, K. Huang, K. Sun — "Descriptor: Power Systems Cascading Failure Analysis Synthetic Dataset (PowerCascade)," IEEE Data Descriptions, 2025	Provide a public, synthetic benchmark dataset for studying and modeling cascading failures in power systems	Data-descriptor paper: generation of synthetic cascading-failure scenarios (likely via simulation of contingencies/outage propagation) with documented structure/format for reproducible research.	Fills a data-availability gap for ML/graph-based cascading-failure research, enabling standardized benchmarking across future studies.
5	K. M, M. P. M, R. M, J. S, S. V. D — "Analysis of Synchrophasor Measurements with Artificial Intelligence Techniques," ICEARS 2025	Apply AI/ML techniques to analyze synchrophasor (PMU) measurement data for grid monitoring.	Likely applies classification/anomaly-detection ML models (e.g., neural nets or ensemble methods) to PMU synchrophasor datasets to detect disturbances or classify grid events.	Supports wide-area situational awareness and early anomaly detection in modern power systems using real-time measurement data.
6	S. Sun et al. — "A Graph Embedding-Based Risk Analysis Case Study of a 2000-bus Cyber-Physical Power System," SmartGridComm 2025	Assess cyber-physical risk in a large-scale (2000-bus) power system using graph representation learning.	Uses graph embedding techniques to represent the cyber-physical network topology and quantify vulnerability/risk propagation paths across a large test system.	Demonstrates scalability of graph-embedding risk analysis to realistic large grids, aiding cyber-physical security planning.

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7	B. Cassottana, P. P. Biswas, S. Balakrishnan, B. Ng, D. Mashima, G. Sansavini — "Predicting Resilience of Interdependent Urban Infrastructure Systems," IEEE Access, 2022	Predict resilience (ability to withstand/recover from disruption) of interdependent urban infrastructure systems (e.g., power, water, transport).	Combines interdependency network modeling with a predictive/data-driven resilience metric, tested via simulated disruption scenarios across coupled infrastructure networks.	Provides urban planners/operators with a predictive tool to prioritize interventions that strengthen cross-sector infrastructure resilience.
8	S. Roy, A. Sen — "A Self-Updating K-Contingency List for Smart Grid System," CCWC 2021	Maintain an up-to-date list of the K most critical (vulnerable) contingencies in a smart grid as system conditions change.	Develops an algorithm that dynamically recomputes/updates the K-contingency list in response to grid-state changes, rather than relying on a static pre-computed list.	Improves real-time grid security assessment by keeping vulnerability rankings current, reducing risk of missed critical contingencies.
9	S. Roy, A. Sen — "Identification of the K-most Vulnerable Entities in a Smart Grid System," CommNet 2020	Identify the K most vulnerable components (nodes/lines) whose failure would cause maximum disruption to a smart grid.	Proposes a computational method (graph/network-based vulnerability scoring) to efficiently rank and identify top-K critical entities, likely evaluated on standard test systems.	Precursor to Roy & Sen's 2021 self-updating work; supports targeted hardening/protection of the most critical grid assets under limited resources.

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S.No	Author, Title & Year	Objective	Methodology	Significance
10	M. Rahnamay-Naeini, M. M. Hayat — "Cascading Failures in Interdependent Infrastructures: An Interdependent Markov-Chain Approach," IEEE Trans. Smart Grid, 2016	Model and predict cascading failure dynamics across interdependent infrastructures (e.g., power and communication networks).	Introduces an interdependent Markov-chain framework where the failure state of one infrastructure's components probabilistically influences the other, capturing cross-network cascade propagation.	A foundational, widely-cited stochastic model for interdependent cascading failures; underpins much later work (including papers #3, #6, #14 in this list) on infrastructure interdependency risk.
11	A. Gholami, F. Aminifar, M. Shahidehpour — "Front Lines Against the Darkness: Enhancing the Resilience of the Electricity Grid Through Microgrid Facilities," IEEE Electrification Magazine, 2016	Examine how microgrids can be leveraged to enhance overall grid resilience during major disruptions (e.g., extreme weather, cascading outages).	Conceptual/review-style analysis of microgrid architectures, islanding capability, and control strategies as resilience-enhancing measures, illustrated with real-world resilience events.	Influential magazine article that helped popularize microgrids as a practical, near-term resilience solution for utilities, shaping subsequent grid-modernization policy and research.
12	S. Lee, L. Chen, S. Duan, S. Chinthavali, M. Shankar, B. A. Prakash — "URBAN-NET: A Network-based Infrastructure Monitoring and Analysis System for Emergency Management and Public Safety," IEEE Big Data 2016	Build a system to monitor and analyze interdependent urban infrastructure networks for use in emergency management and public safety.	Develops a network-based platform (URBAN-NET) integrating multiple infrastructure data streams, applying network-science/graph analysis for real-time monitoring and situational awareness	Provides emergency managers a practical decision-support tool combining big-data infrastructure monitoring with network analytics for disaster response.

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S.No	Author, Title & Year	Objective	Methodology	Significance
13	W. Hurst, N. Shone, Q. Monnet — "Predicting the Effects of DDoS Attacks on a Network of Critical Infrastructures," CIT/IUCC/DASC/PICOM 2015	Predict how a DDoS (Distributed Denial-of-Service) cyberattack propagates and impacts a network of interconnected critical infrastructures	Simulation-based approach modeling DDoS attack propagation across an interdependent infrastructure network to estimate cascading service-disruption effects.	Highlights cyber-attack risk (not just physical/natural hazards) as a driver of critical-infrastructure cascading failure, an early contribution linking cybersecurity with infrastructure interdependency modeling.
14	M. Rahnamay-Naeini, Z. Wang, A. Mammoli, and M. M. Hayat, "A probabilistic model for the dynamics of cascading failures and blackouts in power grids," 2012	Model the dynamics of cascading failures and predict the probability of reaching a blackout of arbitrary size from an initial grid state.	Probabilistic model using two key parameters (maximum capacity of failed lines and grid loading ratio); reduces configuration space via equivalence classes based on number of failed lines and their max capacity; validated against Monte Carlo simulations on the IEEE 118-bus system.	Provides a scalable analytical tool for blackout risk prediction; supports design of control and outage management systems for future smart grids by quantifying cascade probability over time.
15	M. Rahnamay-Naeini, Z. Wang, A. Mammoli, and M. M. Hayat, "A probabilistic model for the dynamics of cascading failures and blackouts in power grids," 2011	Model interdependencies among network systems and quantify how contingencies in one infrastructure can weaken or disrupt the safe operation of interconnected systems through failure propagation.	Simulation framework of a failure cascade process; sensitivity analysis of the critical loading value (lower bound of the cascading-failure region) and average cascade size with respect to coupling parameters that define interdependency strength.	Enables definition and prescription of cascade-safe operating margins for interdependent critical infrastructures, supporting more robust design and operation of multi-network systems.

GAPS IDENTIFIED

1

Limited multi-infrastructure modeling — Most existing studies focus on a single sector (mainly power), with little work modeling how failures propagate across interdependent systems like water, transport, and telecom.

2

Difficulty handling heterogeneous data — Infrastructure data comes in very different formats (sensor readings, network topology, weather, load profiles), and few frameworks handle this variety well together.

3

Lack of unified Big Data + AI real-time pipelines — Current approaches rarely combine large-scale data ingestion, graph-based modeling, and AI prediction into one end-to-end real-time framework — most are isolated academic experiments.

Problem Statement

A single power-grid line failure can cascade into multi-sector outages, yet current models remain static and siloed. This project builds a graph-based AI framework on the IEEE 39-bus system, simulates cascades to train ML/GNN models that predict failure paths and risk in real time, and delivers insights via a decision-support dashboard—with future expansion to interdependent infrastructures.

Dataset

1. IEEE 39-bus system (pandapower.networks.case39())

- **Bus table — 39 rows × 8 columns**
  Bus ID, Bus Type, Pd, Qd, Gs, Bs, Vm, Va, Base kV, Vmax, Vmin

- **Transmission Line table — 35 rows × 16 columns**
  From Bus, To Bus, R, X, B, Rate A, Rate B, Rate C, Tap, Phase Shift, Status, AngleMin, AngleMax

- **Load table — 21 rows × 13 columns**
  Bus, Pd, Qd

- **Generator table — 9 rows × 18 columns**
  Bus, Pg, Qg, Qmax, Qmin, Vg, Status, Pmax, Pmin

link

https://github.com/MATPOWER/matpower/blob/master/data/case39.m
https://github.com/MATPOWER/matpower/blob/master/data/case118.m
https://github.com/MATPOWER/matpower/blob/master/data/case300.m

PLAN OF ACTION

MEMBERS	DESIGNATED WORK	STATUS
VANSH MADHAV	dataset, simulation, and report compilation	FOUND AND STARTED RESEARCH
SUMIT SINHA	ML/GNN development, evaluation, and results.	IN PROGRESS
KAAVYA GUPTA	literature, dashboard, and documentation	IN PROGRESS

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THANK YOU :)